CYS103 Linear Algebra (2026/09)
TUTORIAL 2

Chapter 1: Linear Equations and Linear Transformations

1.3 Solution Sets of Linear Systems

1.4 Linear Transformations

1. Compute the following products in the form of $A\mathbf{x} = \mathbf{b}$. Explain it if the product is undefined.

a. $\begin{bmatrix} 8 & 3 & -4 \\ 5 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} (8\times1)+(3\times1)+(-4\times1) \\ (5\times1)+(1\times1)+(2\times1) \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$

b. $\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix}$ = undefined.

The product is undefined because the number of columns of A is not same as the number of rows of $\vec{x}$. *.

2. In the following statement, (a) does the equation $A\mathbf{x} = 0$ have a nontrivial solution and (b) does the equation $A\mathbf{x} = \mathbf{b}$ have at least one solution for every possible $\mathbf{b}$?

a. A is a 3×3 matrix with three pivot positions.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

(a) It does not have a nontrivial solution.

(b) Yes.

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b. A is a 2×4 matrix with two pivot positions.

(a) $A\vec{x} = 0$ has a nontrivial solution.
(x_1, x_2 are basic V; x_3, x_4 are free V)

(b) $A\vec{x} = \vec{b}$ has a solution for every possible $\vec{b}$ (in $\mathbb{R}^2$)
(because there is a pivot in every row).

3. Determine if the system has a nontrivial solution. Try to use as few row operations as possible.

a.
$$-5x_1 + 7x_2 + 9x_3 = 0$$
$$x_1 - 2x_2 + 9x_3 = 0$$

basic + free variables. → GF

$$\begin{bmatrix} -5 & 7 & 9 & 0 \\ 1 & -2 & 9 & 0 \end{bmatrix} \to \cdots \to \begin{bmatrix} \boxed{1} & -2 & 6 & 0 \\ \boxed{0} & -3 & 39 & 0 \end{bmatrix}$$

(x_1, leading entry, x_2, x_3, c)

x_1 and x_2 are basic variables while x_3 is a free variable. Therefore, the system has a nontrivial solution.

b.
$$2x_1 - 5x_2 + 8x_3 = 0$$
$$-2x_1 - 7x_2 + x_3 = 0$$
$$4x_1 + 2x_2 + 7x_3 = 0$$

$$\begin{bmatrix} 2 & -5 & 8 & 0 \\ -2 & -7 & 1 & 0 \\ 4 & 2 & 7 & 0 \end{bmatrix}$$

Replacement, Scaling ↓ $R2 = R1 + R2$, $R3 = -2R1 + R3$

$$\begin{bmatrix} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 12 & -9 & 0 \end{bmatrix} \xrightarrow{\text{Replacement } R3 = R2 + R3} \begin{bmatrix} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

∴ The system has a nontrivial solution.
(because x_3 is a free variable)

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$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix}$

4. Find the solution sets of the following homogeneous systems in parametric vector form.

a.
$$\begin{aligned} x_1 + 3x_2 + x_3 &= 0 \\ -4x_1 - 9x_2 + 2x_3 &= 0 \\ -3x_2 - 6x_3 &= 0 \end{aligned}$$

basic variables, free variable, REF, nontrivial solution.

$$\begin{bmatrix} 1 & 3 & 1 & 0 \\ -4 & -9 & 2 & 0 \\ 0 & -3 & -6 & 0 \end{bmatrix} \xrightarrow{\text{ERO}} \cdots \rightarrow \begin{bmatrix} 1 & 0 & -5 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

infinite many solutions

Then,

$$\begin{aligned} x_1 - 5x_3 &= 0 \\ x_2 + 2x_3 &= 0 \\ 0 &= 0 \end{aligned} \quad \Rightarrow \quad \begin{aligned} x_1 &= 5x_3 \\ x_2 &= -2x_3 \\ x_3 &\text{ is a free variable} \end{aligned}$$

Therefore,

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$$

$x_3 = 0 \rightarrow \vec{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$x_3 = 1 \rightarrow \vec{x} = \begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}$

$x_3 = 10 \rightarrow \vec{x} = \begin{bmatrix} 50 \\ -20 \\ 10 \end{bmatrix}$

b.
$$\begin{aligned} x_1 + 3x_2 - 5x_3 &= 0 \\ x_1 + 4x_2 - 8x_3 &= 0 \\ -3x_1 - 7x_2 + 9x_3 &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & 3 & -5 & 0 \\ 1 & 4 & -8 & 0 \\ -3 & -7 & 9 & 0 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & 0 & 4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

infinitely many solutions.

Then,

$$\begin{aligned} x_1 + 4x_3 &= 0 \\ x_2 - 3x_3 &= 0 \\ 0 &= 0 \end{aligned}$$

$$\begin{cases} x_1 = -4x_3 \\ x_2 = 3x_3 \\ x_3 \text{ is a free variable} \end{cases}$$

when $x_3 = 0$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0$

when $x_3 = 1$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 1 \end{bmatrix}$

$\vdots$

In parametric vector form: the general solution is $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4x_3 \\ 3x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -4 \\ 3 \\ 1 \end{bmatrix}$. #

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5. With T defined by $T(\mathbf{x}) = A\mathbf{x}$, find a vector $\mathbf{x}$ whose image under T is $\mathbf{b}$, and determine whether $\mathbf{x}$ is unique.

a. $A = \begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & -4 \\ 3 & -5 & -9 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 6 \\ -7 \\ -9 \end{bmatrix}$

Since $T(\vec{x}) = A(\vec{x})$, $A(\vec{x}) = \vec{b} \iff T(\vec{x}) = \vec{b}$.

By $T(\vec{x}) = \vec{b}$, the augmented matrix is:

$$\begin{bmatrix} 1 & -3 & 2 & 6 \\ 0 & 1 & -4 & -7 \\ 3 & -5 & -9 & -9 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

an augmented matrix

a reduced augmented matrix by using ERO

Thus, the vector $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ -3 \\ 1 \end{bmatrix}$ and this solution is unique.

↓ (exactly one solution).

b. $A = \begin{bmatrix} 1 & -5 & -7 \\ -3 & 7 & 5 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$

Since $T(\vec{x}) = A(\vec{x})$, $A(\vec{x}) = \vec{b}$ same as $T(\vec{x}) = \vec{b}$.

$T(\vec{x}) = \vec{b}$ can be expressed in an augmented matrix:

$$\begin{bmatrix} 1 & -5 & -7 & -2 \\ -3 & 7 & 5 & -2 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

Then,

$$\begin{aligned} x_1 + 3x_3 &= 3 \\ x_2 + 2x_3 &= 1 \end{aligned}$$

$$\begin{cases} x_1 = 3 - 3x_3 \\ x_2 = 1 - 2x_3 \\ x_3 \text{ is a free variable.} \end{cases}$$

→

Thus,

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 - 3x_3 \\ 1 - 2x_3 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ -2 \\ 1 \end{bmatrix}$$

The system has no unique solution ※ but infinitely many solutions.

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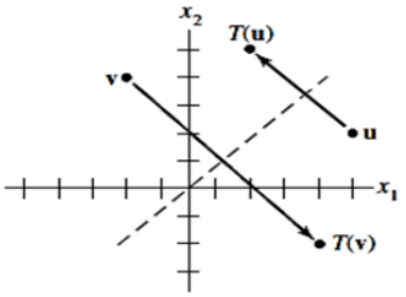
6. Use a rectangular coordinate system to plot $\mathbf{u} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, and their images under the given transformation T. Describe geometrically what T does to each vector $\mathbf{x}$ in $\mathbb{R}^2$.

a. $T(\mathbf{x}) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Images :

$T(\vec{u}) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

$T(\vec{v}) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$



A reflection through the line $x_2 = x_1$.

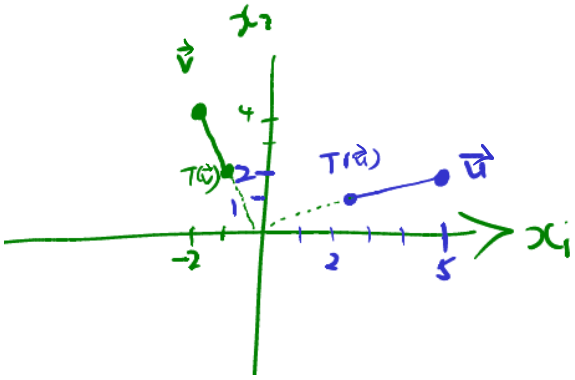
b. $T(\mathbf{x}) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

images

$T(\vec{u}) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 1 \end{bmatrix}$

image

$T(\vec{v}) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$



A contraction by the factor 0.5 #

7. Show that the transformation T defined by $T(x_1, x_2) = (4x_1 - 2x_2, 3|x_2|)$ is not linear.

Property : $Tc(\vec{u}) = cT(\vec{u})$

Let $x_1 = 0$, $x_2 = 1$, and a scalar $c = -1$.

$T(x_1, x_2) = T(0, 1) = (4(0) - 2(1), 3|1|) = (-2, 3)$

$T(-1(x_1, x_2)) = T(-1(0, 1)) = T(0, -1) = (4(0) - 2(-1), 3|-1|) = (2, 3)$

$(-1)T(x_1, x_2) = (-1)T(0, 1) = (-1)(-2, 3) = (2, -3)$

$Tc(\vec{u}) \neq cT(\vec{u})$

Therefore, the transformation T is not linear. because

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8. Let $T: \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation such that

$$T(x_1, x_2) = (x_1 - 2x_2, -x_1 + 3x_2, 3x_1 - 2x_2).$$

Find $\mathbf{x}$ such that $T(\mathbf{x}) = (-1, 4, 9)$.

$$T(\vec{x}) = \begin{bmatrix} x_1 - 2x_2 \\ -x_1 + 3x_2 \\ 3x_1 - 2x_2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A(\vec{x})$$

Since $A\vec{x} = \vec{b}$, $T(\vec{x}) = \vec{b}$ and the augmented matrix is :

$$\begin{bmatrix} 1 & -2 & -1 \\ -1 & 3 & 4 \\ 3 & -2 & 9 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

(x_1, x_2)

augmented matrix

Reduced augmented matrix by using ERO

Then,

$x_1 = 5$

$x_2 = 3$.

$\vec{x}$ in $\mathbb{R}^2$

Thus,

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

#

9. Determine if the following linear transformation is one-to-one. Justify the answer.

$$T(x_1, x_2, x_3, x_4) = (0, x_1 + x_2, x_2 + x_3, x_3 + x_4)$$

write it into a coefficient matrix.

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(so that 0's row is in the last row)

Then, interchange the rows :

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

only 3 pivots

Thus, the transformation T is NOT one-to-one because matrix A does not have a pivot in each row.

(it has a nontrivial solution : x_1, x_2, x_3 : basic variables , x_4 : free variable).

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Application (using small data):

Table 1 presents three ingredients included in the diet, together with the amounts of certain nutrients provided per 100grams of each ingredient.

TABLE 1 The Cambridge Diet

Nutrient	Amounts (g) Supplied per 100 g of Ingredient: Nonfat milk	Soy flour	Whey	Amounts (g) Supplied by Cambridge Diet in One Day
Protein	36	51	13	33
Carbohydrate	52	34	74	45
Fat	0	7	1.1	3

Find some combination of nonfat milk, soy flour, and whey to provide the exact amounts of protein, carbohydrate, and fat supplied by the diet in one day.

$$\begin{matrix} & x_1 & x_2 & x_3 & c \\ R1 \\ R2 \\ R3 \end{matrix} \begin{bmatrix} 36 & 51 & 13 & 33 \\ 52 & 34 & 74 & 45 \\ 0 & 7 & 1.1 & 3 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 0 & 0.277 \\ 0 & 1 & 0 & 0.392 \\ 0 & 0 & 1 & 0.233 \end{bmatrix}$$

Therefore, the diet requires:

0.277 units of nonfat milk,
0.392 unit of soy flour, and
0.233 unit of whey

in order to provide the desired amounts of protein, carbohydrate, and fat. ∎